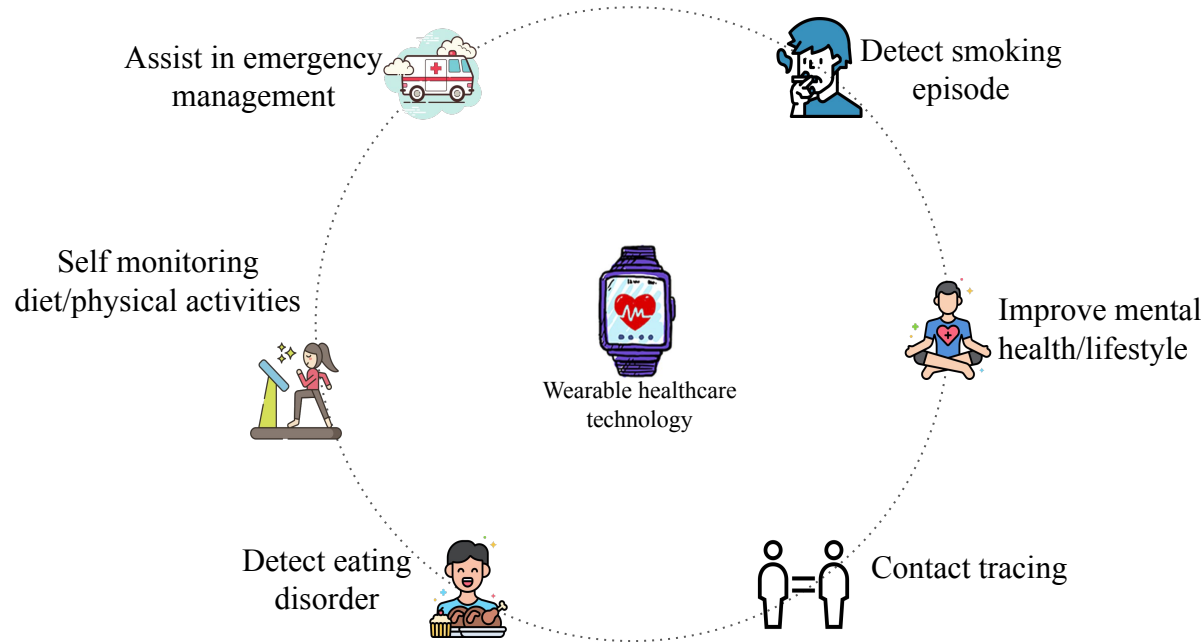


A Secure mHealth Infrastructure for Real-Time Data transfer with Fine-grained Access Control

Saurab Dulal, Nasir Ali, Suravi Regmi, Adam Thieme, Lan Wang
University of Memphis
Tianyuan Yu, Siqi Liu, Lixia Zhang
UCLA

Introduction

- Wearable devices are growing exponentially, e.g., used for health/wellness
- Mobile Health (mHealth) data produced by these devices can do wonders
 - Diagnosis, therapeutics, real-time medical interventions

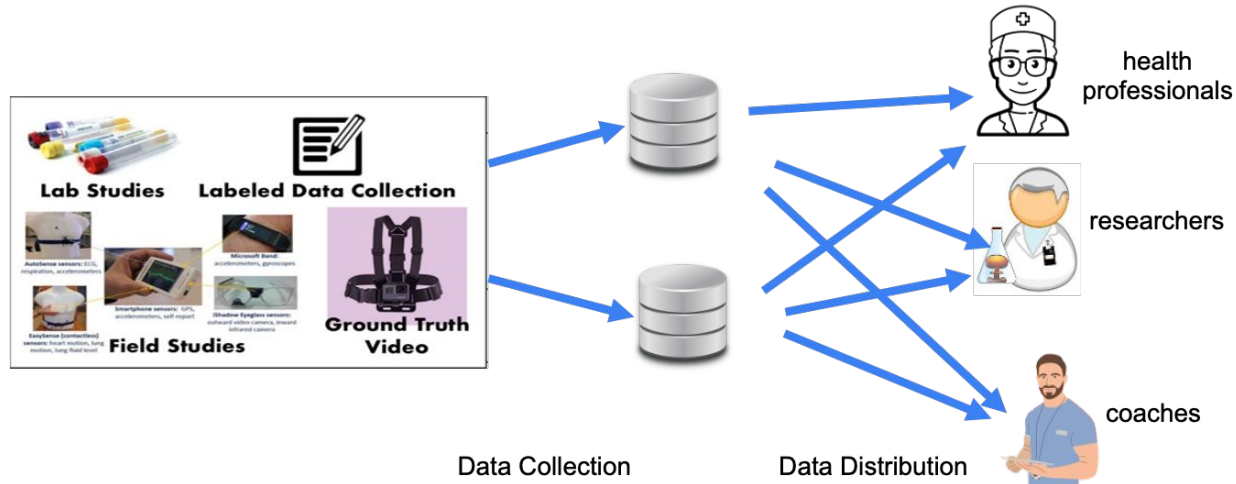


Challenges in Collecting and Sharing mHealth Data

- Highly diverse data
- Privacy and legal concerns

Maintaining Security and privacy is non-trivial when distributing data generated at a high rate and from several sources [9]

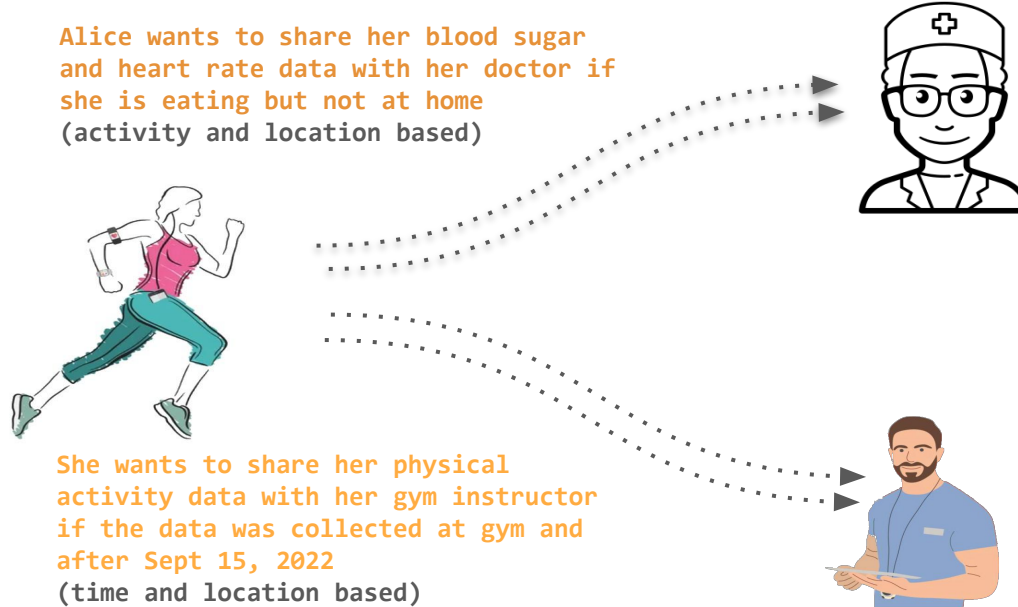
Goals



- An end-to-end data collection and distribution system that can support
 - Complex heterogeneous mHealth data
 - Large-scale and real-time data distribution
- Achieve end-to-end data security
 - Trust between different parties (consumer, producer, operator) involved in the system
 - Fine grained contextual data access control based on policies defined by data the owners

Contextual Access Control

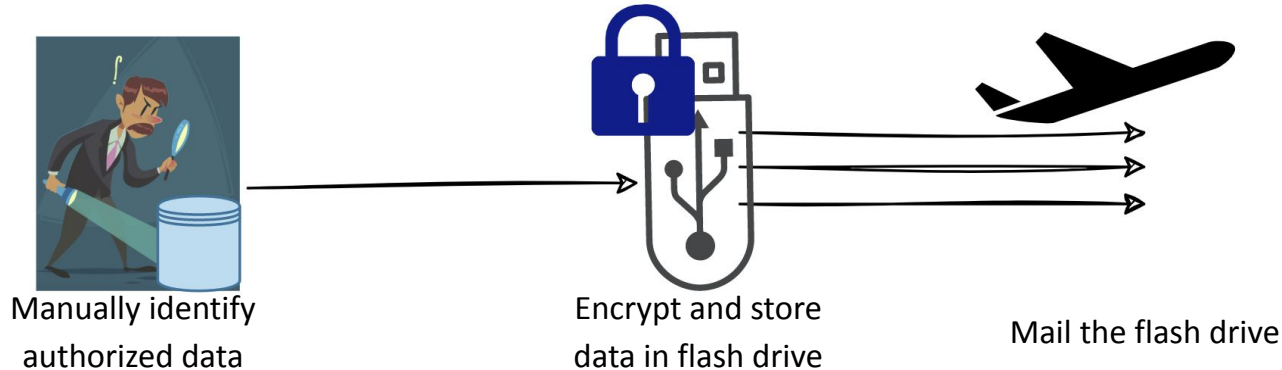
Data sharing based on the context of the data



Limitations of Existing mHealth Systems

MD2K's [6] mHealth research cyberinfrastructure

- Researchers and other users cannot receive real-time data from study participants.
- Limited access control



→ **Slow process, high overhead, and subject to human error** ←

Other “State-of-the-Art ” systems also have several limitations

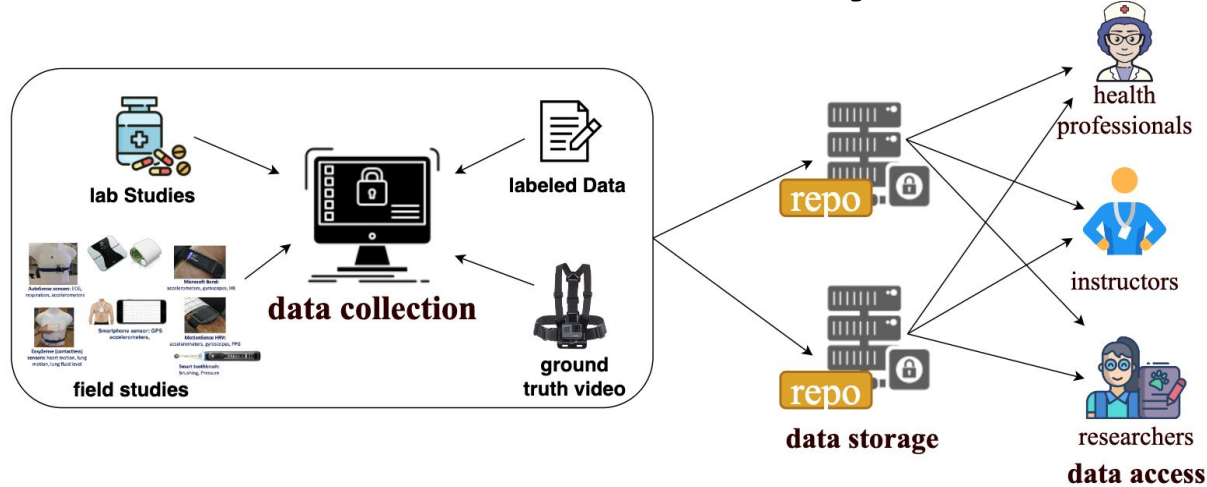
mHealth System	High-Frequency Data	Real-time Data Distribution	Access Control	Contextual Access Control
Microsoft Azure FHIR [1]	N	Y*	Y	N
NIMA [2]	N	N	Y	N
RADAR [3]	N	Y*	Y	N
Adaptive MapReduce [4]	Y	Y	N	N

* partially supported

Contextual Access Control: Data sharing based on the context (location, activity, time) of the data

e.g. data generated at home and when smoking

Use NDN to Build a Secure mHealth System



Benefits

> Data-centric security

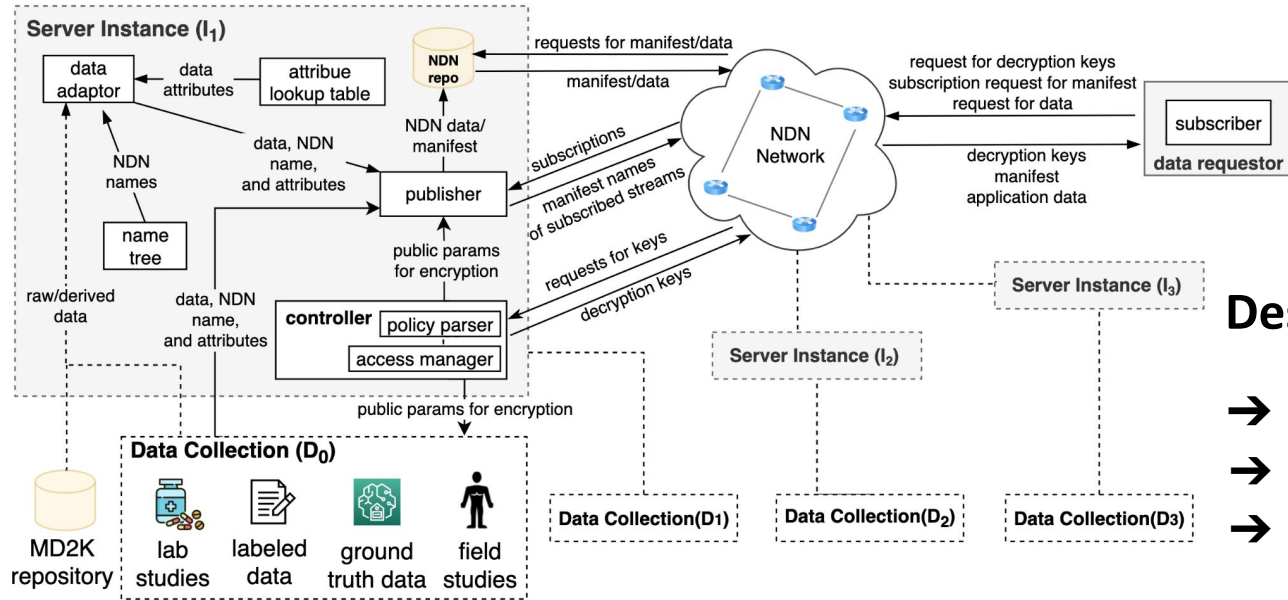
- Key management
- End-to-end protection: data encrypted in transit and in storage
- Fine grained access control using hierarchical/semantic naming

> Real-time data sharing using Sync

> Large scale data distribution

> Persistent storage with NDN Repository

System Overview



Design Elements

- Naming Scheme
- Trust Model
- Access Control
 - ◆ Policy specification
 - ◆ Attribute-based access control
- Publisher Subscriber API

Naming Scheme

(Goal: Design a hierarchical, flexible, and generalizable naming for mHealth system)

- Existing MD2K stream naming (loosely defined)

<sensor>-<namespace>-<device-name>-<attachment>
(e.g. ACCELEROMETER--org.md2k.autosenseble--AUTONSENSE_BLE--CHEST)

- NDN hierarchical naming (figure →)
based on semantic and physical properties

/<data-prefix>/<study-name>/.....
e.g. /org/md2k/mperf/dd40c/phone/gyro/DATA/1492637493876

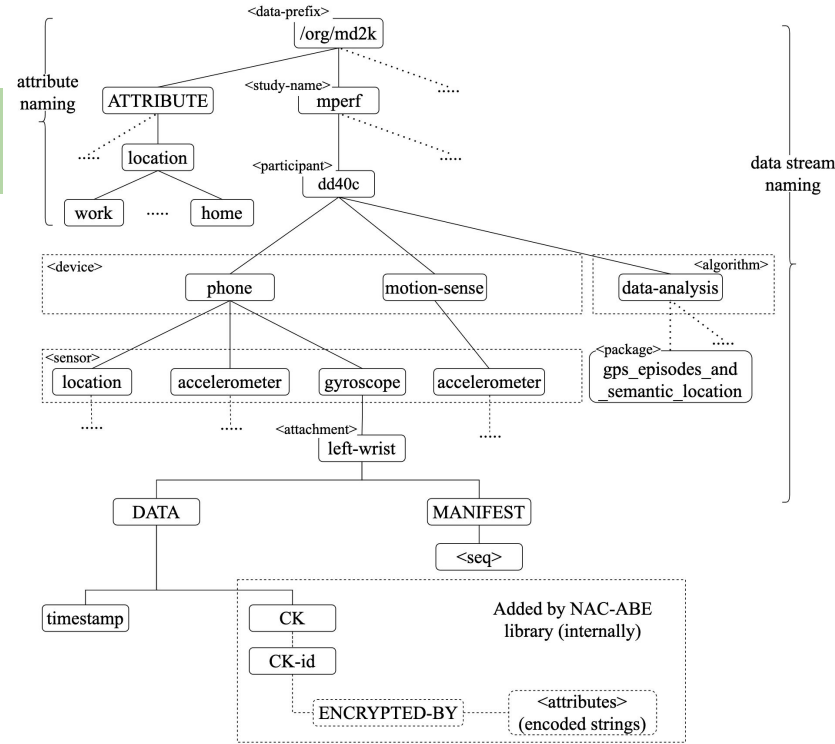
(stream naming)

/<stream name>/DATA/<timestamp>
e.g. /org/md2k/mperf/dd40c/phone/gyro/DATA/1492637493876

(data naming)

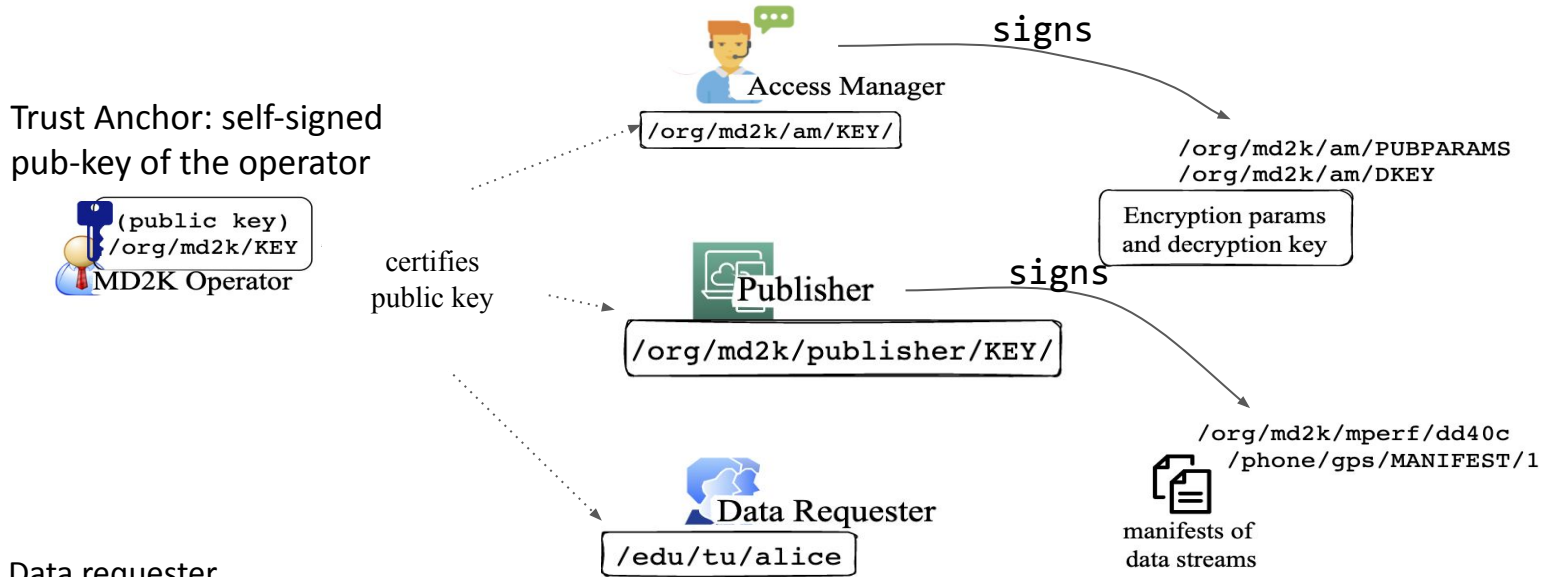
Who else gets a name?

- ATTRIBUTES, MANIFESTS, KEYS and so on



Trust Model

(Goal: Establish trust between different entities in the system)



- Data requester
 - Named under participating organizations' prefixes (e.g. `/edu/campus`)
 - Obtains public key cert from Trust Anchor using NDN CERT
 - Obtains and verifies decryption key (DKEY) from Access Manager
 - Access Manager verifies the requester's public-key cert (signed by trust anchor)

Access Control

(Goals: Provide fine grained access based on attributes/context of the data and **Access Control Policies**)

Design

- Use known data attributes for encryption, Attribute Based Encryption (ABE)
- Accessors (consumers) are unknown during encryption
- Use Key-Policy ABE
 - Encrypt without knowing Access Control Policies
 - Avoid re-encryption when policies changes

Note: we use ABE to encrypt Content Keys (symmetric keys)

- Facilitate faster encryption/decryption of larger objects
- Re-encryption is only CK re-encryption

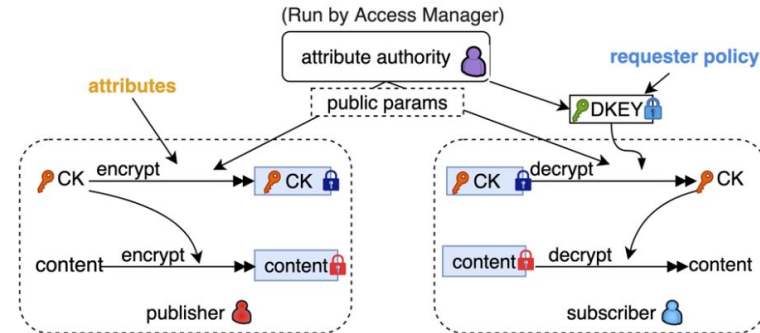


Figure 4: Use of NAC-ABE in mGuard

Access Control Policies

(**Goal:** limit data access to only authorized users based on terms of use and privacy risks)

Default access → Data owner

- Explore their data
- Evaluate privacy concerns

General access → Policies for users

- Participants, Coordinators, Administrators

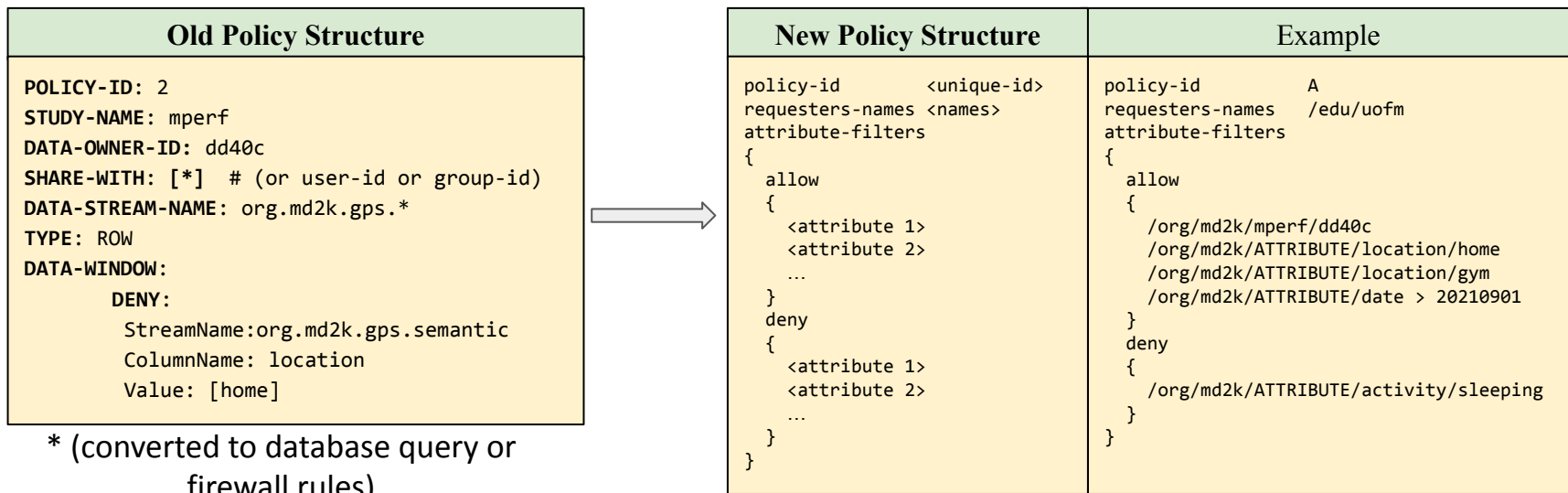
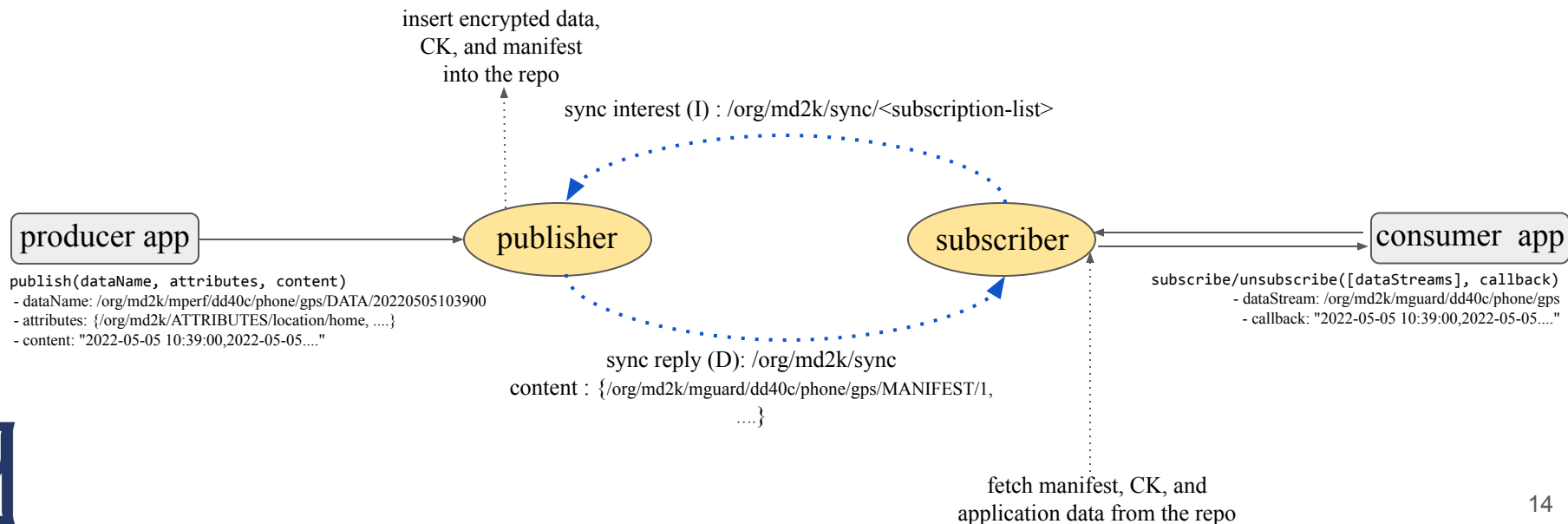


Fig: Evolution of Access Control Policy

Publisher Subscriber API

- Builds off NDN Sync (i.e. PSync [7])
- Security is embedded inside the API
- Authorizes subscribers according to Access Control Policies
- Provides real-time notification to subscriber
- Guarantees the reception of the latest, verified data



Publisher Subscriber API: Manifest

- Signing/Verifying millions of data packets using public-key crypto is computationally expensive
- Publishing individual object names through PSync lead to excessive Sync message overhead, high delays, losses
- **Solution: Manifests**
 - Naming: <data stream name>/MANIFEST/<seq-num>
 - Benefits
 - Bulk verification
 - Application Names → Sequential Names

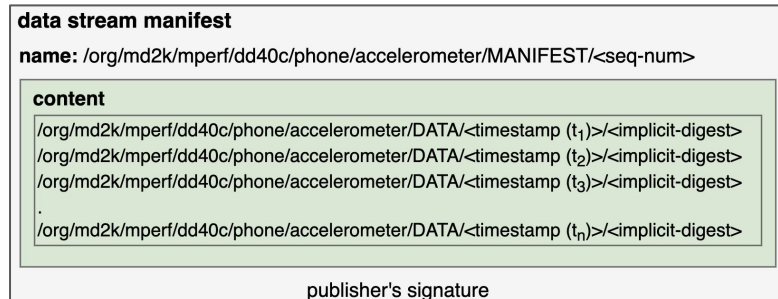


Fig: Manifest Data Format

Implementation

Implementation (open source library)

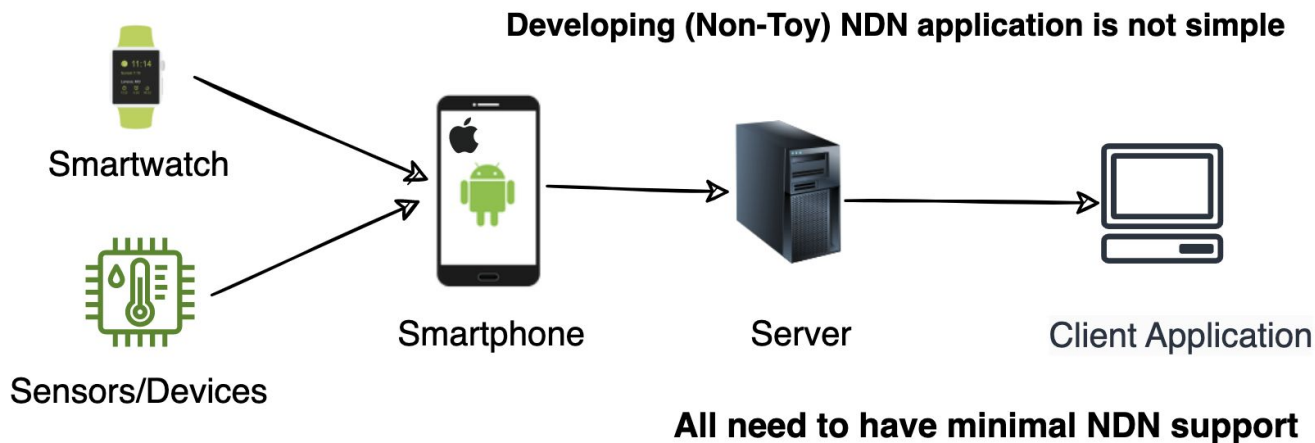
- C++
- High-level pub-sub API
- Uses ndn-cxx, PSync, NDN Python Repo, and NAC-ABE

Source: <https://github.com/dulalsaurab/new-mguard>

Paper: Dulal, S., Ali, N., Thieme, A. R., Yu, T., Liu, S., Regmi, S., Zhang, L., Wang, L. (2022, September). Building a secure mhealth data sharing infrastructure over ndn. In *Proceedings of the 9th ACM Conference on Information-Centric Networking* (pp. 114-124).

Open Issues/Future Work

- OpenABE is not actively maintained
 - NAC-ABE cannot rely on it
 - Lack of negative (NOT) operator support
- KP-ABE approach has a scalability issue
 - Large decryption key for policies with large number of attributes
- End-to-end data privacy



References

- [1] Heather Jordan Cartwright. FHIR Server for Azure: An open source project for cloud-based health solutions. <https://cloudblogs.microsoft.com/industry-blog/health/2018/11/12/fhir-server-for-azure-an-open-source-project-for-cloud-based-health-solutions>.
- [2] Aledavood, T., Hoyos, A. M. T., Alakörkkö, T., Kaski, K., Saramäki, J., Isometsä, E., and Darst, R. K. Data collection for mental health studies through digital platforms: requirements and design of a prototype. *JMIR research protocols* 6, 6 (2017), e110.
- [3] Ranjan, Y., Rashid, Z., Stewart, C., Conde, P., Begale, M., Verbeeck, D., Boettcher, S., Dobson, R., Folarin, A., Consortium, R.-C., et al. Radar-base: Open source mobile health platform for collecting, monitoring, and analyzing data using sensors, wearables, and mobile devices. *JMIR mHealth and uHealth* 7, 8 (2019), e11734.
- [4] Zhang, F., Cao, J., Khan, S. U., Li, K., and Hwang, K. A task-level adaptive mapreduce framework for real-time streaming data in healthcare applications. *Future generation computer systems* 43 (2015), 149–160.
- [5] Saurab Dulal, Nasir Ali, Adam Robert Thieme, Tianyuan Yu, Siqi Liu, Suravi Regmi, Lixia Zhang, and Lan Wang. 2022. Building a secure mHealth data sharing infrastructure over NDN. In *Proceedings of the 9th ACM Conference on Information-Centric Networking (ICN '22)*. Association for Computing Machinery, New York, NY, USA, 114–124. <https://doi.org/10.1145/3517212.3558091>
- [6] MD2K: Center of excellence for mobile sensor data-to-knowledge. MD2K website, <http://md2k.org>.
- [7] Zhang, Minsheng, Vince Lehman, and Lan Wang. "Scalable name-based data synchronization for named data networking." *IEEE Infocom 2017-IEEE Conference on Computer Communications*. IEEE, 2017.
- [8] Zhang, Zhiyi, et al. "An overview of security support in named data networking." *IEEE Communications Magazine* 56.11 (2018): 62-68.
- [9] De Michele, R., and Furini, M. *IoT healthcare: Benefits, issues and challenges*. In *Proceedings of the 5th EAI international conference on smart objects and technologies for social good* (2019), pp. 160–164.

Thank you !

Saurab Dulal
sdulal@memphis.edu